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GPHY488 Jonathan Batchelor

Lab Project 1: Urban Camping

Summarization of Notes and Project:

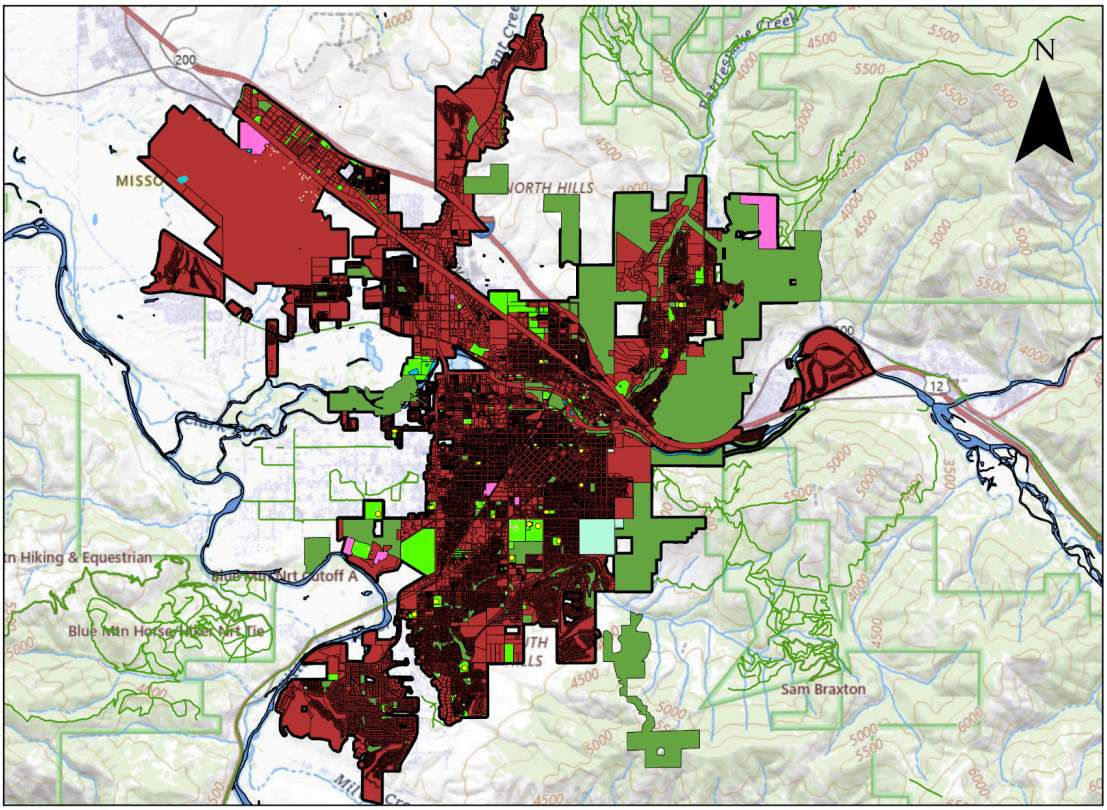
Our team was able to map Missoula's current urban camping restriction. Ethan Jones and Callie Rommel were mainly in charge of mapping and data cleaning, while Morgan Kelly and Sofia Lopez were also in charge of finding data, the dashboard and cleaning. We had a group chat where we assigned separate duties to complete on this project outside of lab time when necessary. This includes buffer zones around residential buildings (100 feet), business entrances (100 feet), schools and daycare facilities (300 feet), waterways (50 feet from the high water mark), shelters, parks and conservation lands, pathways, trails and roadways. We projected all layers into the Montana State Plane (EPSG 2256), otherwise North American Datum 1983 (4269) to ensure consistency and spatial accuracy.

However, we were not able to fully map every restriction. We decided to copy the rules and regulations of Missoula Camping perimeters into a shared document, and color-coded it by what we had done and been completed/exported, what we needed to still work on or needed to find or be expanded on, and what was still in the process or needed to be started. The most significant barrier was being unable to map areas temporarily designated as no-camping zones due to environmental degradation or because of public health safety concerns.

We were unable to spatially define these temporary designated no-camping zones as they are periodically defined by the mayor or posted on the city's camping webpage. Another barrier was being unable to map safe parking designations. The car camping ordinance allows only one permitted vehicle per city block, but it came with broad and conditional stipulations making it difficult to properly translate it into a clear spatial layer. The skills required to complete this map included spatial analysis, buffering, clipping, projection management, data cleaning and policy interpretation and research. This suggests that Missoula's urban camping policy is highly technical and layered and requires significant geographic analysis to fully understand and implement. Lastly, to get one of our layers to save within our dashboard to be able to show our layer shelter locations when you click on the symbology.

Dashboard link: Urban Camping Project_FINAL - Overview

Missoula City Urban Camping



Scale: 1:165,000

Legend

- Missoula City Limits
- Parks in Missoula
- Trails
- Waterways in Missoula
- FederalLand
- Missoula City Property
- Private Property
- Montana State Land
- Residences Buffer
- Trails Buffer
- Buisnesses buffer
- Schools Buffer
- Waterways Buffer

0 0.75 1.5 3 Miles

Authors: Callie Rommel, Ethan Jones, Morgan Kelly, Sofia Lopez

Spatial Reference
GCS: GCS WGS 1984

Links for Data used throughout the process:

Missoula Maps
Missoula Maps
TNM Download v2
Missoula County Parks, Trails, and Open Lands
County Maps | Missoula County, MT
Missoula County Open Data
Data List - Home
Montana Geographic Information
Data.gov Home - Data.gov
GIS & Mapping Search - LawInfoPedia
Copernicus Browser
GIS and GPS Downloadable Data
Free GIS shapefiles - Simply Geography